ENTERPRISE DATABASE CLOUD BACKUP & MAINTENANCE

Administrator Installation, Provisioning, Disaster Recovery & Operations Manual · v5.0.0

Document Version	Target Audience	Classification	Release Track
v5.0.0 Production	IT Administrators / DevOps	RESTRICTED (Internal IT Only)	Zero-Trust Architecture

> **[SECURITY NOTICE]** CONFIDENTIAL & RESTRICTED: This package and documentation must ONLY be used on the central IT Administrator workstation. It contains master encryption tools, private signing keys, disaster recovery decryptors, and customer provisioning logic. DO NOT distribute this package or any of its tools to customer servers.

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1. Executive Overview & Zero-Trust Architecture

The Enterprise Database Cloud Backup & Maintenance Suite provides autonomous, cryptographically verified database protection and maintenance across 70+ customer SQL Server deployments. Backups are encrypted 100% client-side with AES-256-GCM and dual RSA-4096 envelope encryption before uploading via a serverless Google Apps Script broker to Google Drive. The architecture strictly isolates administrative powers from customer servers.

Component	Location	Security Role & Guarantees
Admin Installation Package	IT Admin Workstation	Houses customer provisioning scripts, bundle signing keys (Ed25519), and offline recovery decryptors.
Client Installation Package	Customer Server	Holds ONLY public keys (backup_public.pem). Can encrypt backups but CANNOT decrypt them. Zero Python.
Google Apps Script Broker	Google Cloud / Workspace	Stateless serverless gateway. Uploads chunks to Google Drive and logs telemetry to Google Sheets without GCP billing.

2. Admin Workstation Prerequisites & 1-Click Setup

Follow these exact steps to prepare your administrative machine:

1. Step 1: Verify Python 3.9 or newer is installed on your Windows workstation. Ensure 'Add Python to PATH' was checked during installation.
2. Step 2: Navigate to the Admin_Installation_Package root folder.
3. Step 3: Double-click Setup_Admin_Environment.bat (or execute it in Command Prompt).
4. Step 4: The automated script creates an isolated virtual environment (.venv), upgrades pip, installs cryptography, requests, and google-api-client, and validates imports.
5. Step 5: You will see the green confirmation banner: 'ADMIN ENVIRONMENT IS 100% READY!'.

```
cd C:\Admin_Installation_Package
Setup_Admin_Environment.bat
```

3. Google Apps Script Broker & Cloud Storage Deployment

The Google Apps Script Master Broker (Code.gs) acts as the secure, zero-trust serverless relay between all 70+ client servers and your central Google Drive & Sheets repositories.

6. Step 1: Open Google Drive in your web browser using your administrative account.
7. Step 2: Create a parent folder for all backups, e.g., 'Enterprise Customer Backups'. Note its Folder ID from the URL (the string after /folders/).
8. Step 3: Create a master Google Sheet named 'Application report'. Note its Spreadsheet ID from the URL (the string between /d/ and /edit).
9. Step 4: Go to https://script.google.com and click 'New project' (or open your spreadsheet -> Extensions -> Apps Script).
10. Step 5: Replace all code in Code.gs with the complete content of 3_Cloud_Broker_Deployment\Code.gs.
11. Step 6: Update lines 12-13 with your Spreadsheet ID and Parent Drive Folder ID.
12. Step 7: In the left sidebar, click Project Settings (Gear icon) -> check 'Show appsscript.json manifest in editor' -> paste the content from 3_Cloud_Broker_Deployment\appsscript.json.
13. Step 8: Click the Deploy button -> New deployment -> Select type: Web App.
14. Step 9: Configure: Execute as: 'Me' (your admin account), Who has access: 'Anyone'. Click Deploy.
15. Step 10: Copy and save the resulting Web App URL (starts with https://script.google.com/macros/s/.../exec).

> **[CRITICAL DEPLOYMENT SETTING]** Always select 'Anyone' for 'Who has access'. This enables zero-trust client servers to upload encrypted ciphertext and post telemetry via HTTPS without requiring client Google account credentials.

4. Auto-Generating 70+ Customer Tabs in Google Sheets

To save hours of manual spreadsheet configuration, Code.gs includes an automated initialization routine that instantly generates all 70+ customer tabs in your Master 'Application report' spreadsheet.

16. Step 1: In the Google Apps Script editor, locate the toolbar function dropdown menu (defaults to doGet or doPost).
17. Step 2: Select the function named 'setupApplicationReportTabs'.
18. Step 3: Click the Run button.
19. Step 4: When prompted, grant permissions to allow the script to manage spreadsheet tabs.
20. Step 5: The script executes in ~10 seconds and automatically generates individual formatted tabs for all 70+ customers (adelaideglass, bnd, viridian, etc.) with frozen header rows.

5. Customer Provisioning Workflows (Single & 70+ Batch)

Customer provisioning generates unique 4096-bit RSA keys, seals the configuration in a cryptographically signed bundle.json, and outputs a customer-ready deployment package with zero secret leakage.

Option A: Batch Provisioning All 70+ Customers in One Pass

21. Step 1: Open command prompt in Admin_Installation_Package\1_Customer_Provisioning.
22. Step 2: Verify or edit customer_slugs.txt (one customer per line: slug,Display Name).
23. Step 3: Run: python batch_setup_customers.py
24. Step 4: Paste your Apps Script Broker URL when prompted.
25. Step 5: Enter your Admin Ed25519 signing key path and passphrase.

26. Step 6: Choose whether to use a shared passphrase or individual passphrases for recovery keys.
27. Step 7: The wizard generates keys, signs bundles, and creates customers\<slug>_package for all customers.
28. Step 8: Open customers\enroll_codes_for_sheet.tsv and paste the rows directly into columns A, B, C of your Google Sheet 'Config' tab.

```
cd C:\Admin_Installation_Package\1_Customer_Provisioning
python batch_setup_customers.py
```

Option B: Single Customer Provisioning Wizard

29. Step 1: Run: python setup_new_customer.py
30. Step 2: Enter customer slug (e.g. acme) and Apps Script Broker URL.
31. Step 3: Choose a strong passphrase (14+ characters) for the customer's private keys.
32. Step 4: Provide your Admin Signing Key to sign bundle.json.
33. Step 5: Copy the output row (ENROLL_CODE <code-string> acme) into your Google Sheet Config tab.
34. Step 6: The customer deployment package is saved in customers\acme_package.

6. Air-Gapped Disaster Recovery & SQL Server Restoration

If a customer server suffers ransomware, hardware failure, or corruption, use the air-gapped recovery suite to decrypt the raw database (.bak) and restore it to SQL Server.

Step-by-Step Restoration Runbook:

35. Step 1: Download the target encrypted backup file (.dbk2) from the customer's folder in Google Drive.
36. Step 2: Place the file on your Admin workstation or secure restoration server.
37. Step 3: Double-click 2_Disaster_Recovery\Launch_Recovery_Wizard.bat (or run python decrypt_gui.py).
38. Step 4: Click 'Browse Encrypted File' and select the downloaded .dbk2 file.
39. Step 5: Click 'Browse Private Key' and select the customer's backup_private.pem (or the master escrow_private.pem).
40. Step 6: Enter the key passphrase and choose the output folder for the restored database.
41. Step 7: Click 'Start Decryption & Integrity Verification'. The tool verifies the 128-bit AES-GCM MAC tag and produces the authentic .bak file.
42. Step 8: Open SQL Server Management Studio (SSMS) -> right click Databases -> Restore Database -> Device -> select the decrypted .bak file -> click OK.

```
-- SQL Server T-SQL Fast Restore Script
RESTORE DATABASE [TheDatabase]
FROM DISK = N'C:\Restores\TheDatabase_Restored.bak'
WITH REPLACE, RECOVERY, STATS = 10;
GO
```

7. Complete Step-by-Step Uninstallation & Decommissioning Processes

When an administrator needs to decommission an agent or guide a customer through complete removal, follow these strict uninstallation procedures to ensure zero orphaned scheduled tasks or credential leaks.

Method	Target Environment	Action & Scope
PowerShell Uninstaller (uninstall_agent.ps1)	Server Core / Headless / PowerShell Agent Track	Terminates active processes, deletes all 3 scheduled tasks, shreds token.dpapi with random bytes, deletes scripts and directories.
Windows 1-Click Uninstaller (Uninstall.bat)	Windows Desktop / Full GUI Track	Self-elevating batch uninstaller. Cleans Program Files, ProgramData, registry uninstall keys, and shortcuts across all profiles.
Windows Settings / appwiz.cpl	Standard Windows Control Panel	Invokes registered UninstallString from registry. Silently or interactively executes clean uninstallation.

Admin Decommissioning Checklist:

- 1. Instruct customer to run uninstall_agent.ps1 or Uninstall.bat on their server.
- 2. Open the Master Google Sheet -> go to the 'Tokens' tab -> delete the enrolled row corresponding to the decommissioned server.
- 3. Go to the 'Config' tab -> clear the customer's ENROLL_CODE row.
- 4. In Google Drive, archive or retention-lock the customer's historical backup folder according to company compliance policies.
- 5. Move the customer's offline private keys (<slug>_keys) into long-term encrypted archive vault.

8. Key Management & Long-Term Security Protocols

Key Identifier	Algorithm & Key Size	Storage Policy & Restrictions
Admin Signing Key (admin_ed25519.pem)	Ed25519 (256-bit elliptic curve)	Admin hardware only. Never in cloud-synced folders (OneDrive/Dropbox). Protected by 14+ char passphrase.
Customer Primary Key (<slug>_private.pem)	RSA-4096 (PKCS#8 Encrypted)	Stored offline in 2_Disaster_Recovery\keys. Used for customer restoration requests.

Enterprise Escrow Key (escrow_private.pem)	RSA-4096 (PKCS#8 Encrypted)	Master emergency recovery key. Can decrypt any archive if customer key is lost. Keep in physical offline safe.